

Assume a wheel A rotates about an axle. Without loss of generality, assume the axle is located at $\vec{r}_0 = (0,0)$. Assume there is a wheel around that axle, defined by a parametric curve. Let the parametric curve be given by:

$$\vec{a}(t) = [x(t), y(t)]$$

Assume that $\vec{a}(t)$ is closed, and that $\vec{r}_0 \in I(\vec{a}(t))$. We are going to suppose that the point $\vec{a}(t)$ on Wheel A is the point intersecting with the x-axis. With this, we can rewrite $\vec{a}(t)$ as follows:

$$\begin{aligned}\rho(t) &= \sqrt{(x(t))^2 + (y(t))^2} \\ \theta(t) &= \arctan(y(t), x(t)) \\ \vec{a}(t) &= \rho(t) \begin{bmatrix} \cos(\theta(t)) \\ \sin(\theta(t)) \end{bmatrix}\end{aligned}$$

As this parametric equation still represents a wheel, it must be able to rotate. If we wish to rotate $\vec{a}(t)$ by some angle α about $\vec{r}_0$, we construct the following matrix:

$$\vec{R}(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

Because $\vec{a}(t)$ must always be affixed to the x-axis, as we iterate through all points in $\{t | t \in [0, 2\pi]\}$, we must rotate $\vec{a}(t)$ back to the x-axis by angle $-\theta(t)$.

$$\alpha = \alpha(t) = -\theta(t)$$

For the intersection point of wheel A with the x-axis, across all t , we have:

$$\begin{aligned}\vec{p}_A &= \vec{R}(\alpha)\vec{a}(t) = \vec{R}(-\theta(t))\vec{a}(t) = \begin{bmatrix} \cos(-\theta(t)) & -\sin(-\theta(t)) \\ \sin(-\theta(t)) & \cos(-\theta(t)) \end{bmatrix} \rho(t) \begin{bmatrix} \cos(\theta(t)) \\ \sin(\theta(t)) \end{bmatrix} \\ &= \rho(t) \begin{bmatrix} \cos(-\theta(t)) & -\sin(-\theta(t)) \\ \sin(-\theta(t)) & \cos(-\theta(t)) \end{bmatrix} \begin{bmatrix} \cos(\theta(t)) \\ \sin(\theta(t)) \end{bmatrix} \\ &= \rho(t) \begin{bmatrix} \cos(-\theta(t))\cos(\theta(t)) - \sin(-\theta(t))\sin(\theta(t)) \\ \sin(-\theta(t))\cos(\theta(t)) + \cos(-\theta(t))\sin(\theta(t)) \end{bmatrix} \\ &= \rho(t) \begin{bmatrix} \cos(\theta(t))\cos(\theta(t)) + \sin(\theta(t))\sin(\theta(t)) \\ -\sin(\theta(t))\cos(\theta(t)) + \cos(\theta(t))\sin(\theta(t)) \end{bmatrix} \\ &= \rho(t) \begin{bmatrix} \cos^2(\theta(t)) + \sin^2(\theta(t)) \\ -\sin(\theta(t))\cos(\theta(t)) + \sin(\theta(t))\cos(\theta(t)) \end{bmatrix} \\ &= \rho(t) \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} \rho(t) \\ 0 \end{bmatrix}\end{aligned}$$

If we assume another wheel is located at $\vec{r}_1 = (L, 0)$ without loss of generality and similarly parameterize the second wheel B with a closed curve $\vec{b}(s)$ such that $\vec{r}_1 \in I(\vec{b}(t))$, and make the following two assumptions:

1. The wheels only touch one another at one point, $\vec{a}(t)$.
2. The wheels carry the same tangential velocity as one another, but in opposite directions.

Because both wheels only share one point of contact, being $p(t) = \begin{bmatrix} \rho(t) \\ 0 \end{bmatrix}$, we will define the distance from $\vec{r}_0$ to the point of contact as $\lambda(t) = \rho(t)$, and the distance from the point of contact to $\vec{r}_1$ as $\mu(t) = L - \rho(t)$. This gives $L > \rho(t)$ for all t.

Similarly to $\alpha(t)$, define $\beta(s)$ as the rotation angle of wheel B. We can impose the condition that both tangential velocities must be equal at the point of contact. Define a physical time τ . At the point of contact, the tangential velocity of wheel A can be determined as:

$$\begin{aligned}
 \dot{\vec{p}}_A &= \frac{d}{d\tau}(\vec{R}(\alpha)\vec{a}(t)) = \frac{d}{d\tau}(\vec{R}(\alpha))\vec{a}(t) + \vec{R}(\alpha)\frac{d}{d\tau}(\vec{a}(t)) \\
 &= \frac{d\vec{R}}{d\alpha}\frac{d\alpha}{d\tau}\vec{a}(t) + \vec{R}(\alpha)\frac{d\vec{a}}{dt}\frac{dt}{d\tau} = \frac{d\vec{R}}{d\alpha}\frac{d\alpha}{d\tau}\vec{a}(t) + 0 \\
 &= \frac{d}{d\alpha}\left(\begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix}\right) \cdot \frac{d}{d\tau}(\alpha(t)) \cdot \vec{a}(t) \\
 &= \begin{bmatrix} -\sin(\alpha) & -\cos(\alpha) \\ \cos(\alpha) & -\sin(\alpha) \end{bmatrix} \dot{\alpha}(t)\vec{a}(t) \\
 &= \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\alpha}(t)\vec{a}(t) \\
 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\alpha}(t) \cdot \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} \vec{a}(t) \\
 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\alpha}(t) \cdot \vec{R}(\alpha)\vec{a}(t) \\
 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\alpha}(t)\vec{p}_A \\
 &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\alpha}(t) \cdot \begin{bmatrix} \rho(t) \\ 0 \end{bmatrix} \\
 &= \dot{\alpha}(t) \begin{bmatrix} 0 \\ \rho(t) \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ \dot{\alpha}\rho(t) \end{bmatrix}
 \end{aligned}$$

For wheel B, everything follows the exact same way, except for the fact that $\vec{p}_B = \begin{bmatrix} L - \rho(t) \\ 0 \end{bmatrix}$, since we know that $\vec{r}_0$ and $\vec{r}_1$ are L apart:

$$\dot{\vec{p}}_B = \frac{d}{d\tau}(\vec{R}(\beta)\vec{b}(s)) = \frac{d\vec{R}}{d\beta}\frac{d\beta}{d\tau}\vec{b}(s) + \vec{R}(\beta)\frac{d\vec{b}}{ds}\frac{ds}{d\tau} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\beta} \cdot \vec{R}(\beta)\vec{b}(s)$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \dot{\beta} \begin{bmatrix} L - \rho(t) \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \dot{\beta}(L - \rho(t)) \end{bmatrix}$$

Equating both tangential velocities yields the following equation:

$$\begin{aligned} \begin{bmatrix} 0 \\ \dot{\alpha}\rho(t) \end{bmatrix} &= \begin{bmatrix} 0 \\ \dot{\beta}(L - \rho(t)) \end{bmatrix} \Rightarrow \dot{\alpha}\rho(t) = \dot{\beta}(L - \rho(t)) \Rightarrow \dot{\beta} = \frac{\rho(t)}{L - \rho(t)} \dot{\alpha} \\ \Rightarrow \dot{\beta} &= \frac{\rho(t)}{L - \rho(t)} \cdot \frac{d}{d\tau}(-\theta(t)) \Rightarrow \frac{d\beta}{d\tau} = -\frac{\rho(t)}{L - \rho(t)} \cdot \frac{d\theta}{dt} \frac{dt}{d\tau} \\ \Rightarrow \frac{d\beta}{dt} &= -\frac{\rho(t)}{L - \rho(t)} \cdot \frac{d\theta}{dt} \Rightarrow \beta(t) = -\int_{t_0}^t \frac{\rho(u)}{L - \rho(u)} \cdot \frac{d\theta}{du} du \end{aligned}$$

Now armed with an expression for $\beta(t)$, we can lock in the value for L . We already know that $t \in [0, 2\pi]$. For every full rotation of A, B must rotate k times. This gives us:

$$\int_{t_0}^{t_0+T} \frac{\rho(t)}{L - \rho(t)} \theta'(t) dt = 2\pi k$$

to define L . With $\beta(t)$ as above, we can denote the curve of $b(t)$ as:

$$b(t) = R(-\beta(t)) \begin{pmatrix} \rho(t) - L \\ 0 \end{pmatrix} = \begin{bmatrix} -(L - \rho(t)) \cos \beta(t) \\ (L - \rho(t)) \sin \beta(t) \end{bmatrix}$$